CROP DISEASE DETECTION USING IMAGE CLASIFICATION

Application Development-I Report submitted to JNTU Hyderabad

in partial fulfillment of the requirements for the award of the

degree

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

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An UGC Autonomous Institution

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Maisammaguda, Medchal (Dist), Hyderabad -500100, T.S

OCTOBER 2025

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CERTIFICATE

This is to certify that the Application Development-I entitled **"CROP DISEASE DETECTION USING IMAGE CLASSIFICATION"** has been submitted By K. Manideep (23N31A0456) in partial fulfillment of the requirements for the award of **BACHELOR OF TECHNOLOGY IN ELECTRONICS & COMMUNICATION ENGINEERING**.This record of bonafide work was carried out by them under my guidance and supervision. **The result embodied in this Application Development-I report has not been submitted to any other University or Institute for the award of any degree.**

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DECLARATION

We hereby declare that the Application Development-I titled "**Crop Disease Detection Using Image Classification**" submitted to Malla Reddy College of Engineering and Technology (UGC Autonomous), affiliated to Jawaharlal Nehru Technological University Hyderabad (JNTUH) for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering is a result of original work carried-out in this report. It is further declared that the Application Development-I report or any part thereof has not been previously submitted to any University or Institute for the award of degree or diploma.

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ACKNOWLEDGMENTS

The Application Development-I work carried out by our team in the Department of Electronics and Communication Engineering, Malla Reddy College of Engineering & Technology, Hyderabad. This work is original and has not been submitted in part or full for any degree or diploma of any other university. We wish to acknowledge our sincere thanks to our project guide **K.D.K. Ajay**, Assistant Professor of Electronics & Communication Engineering Department for formulating the problem, analysis, guidance and his continuous supervision during the project work. We acknowledge our sincere thanks **to Dr. Vaka Murali Mohan**, Principal, MRCET**. Dr. K Mallikarjuna Lingam**, Professor and Head of the Electronics & Communication Engineering Department, for their kind cooperation in making this Application Development-I work a success. We extend our gratitude to **Sri. Ch. Malla Reddy**, Founder-Chairman and **Sri. Ch. Mahender Reddy**, Secretary, MRCET and **Sri. P. Praveen Reddy** Director, MRGI for their kind cooperation in providing the infrastructure for successful completion of our Application Development-I work. We convey our special thanks to the Entire Teaching faculty and non-teaching staff members of the Electronics & Communication Engineering Department for their support in making this project work a success.

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LIST OF ACRONYMS

ABBREVIATION	FULL FORM
KNN	K-Nearest Neighbour
SVM	Support Vector Machine
CNN	Convolutional Neural Network
GPDCNN	Global pooling dilated CNN
MCNN	Multilayer convolutional neural network
PSO	Particle Swarm Optimization
DCNN	Deep Convolution Neural Network
CCM	Color Co-occurrence Matrix
GLCM	Gray Level Co-occurrence Matrix
MER	Minimum Enclosing Rectangle
DWT	Discrete Wavelet Transform
SIFT	Scale Invariant Feature Transform
HOG	Histogram of Oriented Gradients
STL-10	Self-Taught Learning 10
ResNet	Residual Network
HSV	Hue Saturation Value
RGB	Red, Green, Blue
HDF	Hierarchical Data Format
AI	Artificial intelligence
MDP	Markov decision process
ML	Machine Learning
HTTP	HyperText Transfer Protocol

WSGI	Web Server Gateway Interface
PEP	Python Enhancement Proposal
BSD	Berkeley Source Distribution
AQR	Applied Quantitative Research
DBSCAN	Density-based spatial clustering of applications with noise
API	Application programming interface
RF	Random Forest
LR	Logistic Regression
VGG	Visual Geometry Group
JPEG	Joint Photographic Experts Group
BMP	Bitmap

ABSTRACT

This crop disease detection application leverages machine learning models to classify images of crop leaves and predict the health status of the crop. Built with TensorFlow and Streamlit, the app provides an intuitive interface where users can upload an image, which is then preprocessed and fed into an ensemble model composed of Xception and DenseNet architectures. The model, trained on disease classification, delivers predictions indicating whether the crop is healthy or diseased. The code includes efficient caching of model weights, image cleaning functions, and clear visualization of results with progress tracking. The application hides Streamlit's default menu for a more focused user experience, aiming to be accessible and useful for agricultural diagnostics and crop health monitoring.

CHAPTER 1
INTRODUCTION

1.1 PROJECT OBJECTIVE:

The primary objective of crop disease detection is to identify and diagnose crop diseases accurately and quickly. This is important to prevent the spread of the disease and to minimize crop loss. Early detection of crop diseases is crucial to prevent the spread of the disease and to minimize crop damage. Disease detection using Machine Learning Algorithms should be reliable and produce consistent results.

1.2 CROP LEAF DISEASE:

A disturbance in the stock situation of a herb that destroys or alters crucial. All types of herbs, both unbroken and educated, are susceptible to illness. Each family is prone to specific diseases, but each of these is relatively rare. The incidence and prevalence of crop illness vary from prime to prime, pivoting on the company of pathogens, territory conditions, and the supply and cultivars grown. Some crop varieties are particularly susceptible to disease epidemics, while others are more resilient. See also list of herb illness.

- **Definitions of crop disease**

Generally, when crops are consistently disturbed, they might catch illness. pathogens that result in abnormal physiological processes that disrupt the normal structure, growth, function, or other activities of the crop. The essential physiological or biochemical processes of the crop are disrupted, which results in the typical diseased states or symptoms. Depending on whether the primary cause of the disease is infectious or non- infectious, crop diseases may be broadly categorized. Infectious crop diseases are caused by pathogens such as fungi, bacteria, mycoplasma, viruses, viroids, nematodes, or parasitic flowering crop s. Within or on a host, infectious organisms can grow and spread from one vulnerable host to another. Unfavorable growth circumstances, including as excessive temperatures, unfavorable moisture-oxygen ratios, soil and air pollutants, and an abundance or shortage of vital minerals, are the root causes of non-infectious crop illnesses.

- **Temperature**

All pathogens have an optimum growth temperature. In addition, the optimal temperature may differ slightly due to different stages of fungal growth, such as the production of spores (reproductive units), their germination, and the growth of the mycelium (the filamentous body of the fungus). Warehouse conditions for certain berry, greens, and nest produce are manipulated to control fungi and bacteria that cause storage spoilage, so long as the febricity does not alter the quality of the produce. Aside from limited dip protection, there is little you can do to control the temperature in your field, but you can adjust the temperature in your greenhouse to reduce disease outbreaks. Combined with high humidity conditions, it allows the development of diseases such as vine downy mildew (Pseudoperonospora cubensis), lima bean (Phytophthora phaseoli) and late blight on potatoes. These include precision tomato (Phytophthora infestans), sugar beet leaf spot . However, temperature effects can mask the symptoms of certain aggressive and mycoplasmal diseases, making them more onerus to detect.

- **Relative humidity**

Relative moisture is exact important for the germination of fungal spores and the evolution of warehouse rot. Rhizopus stolonifer of sweet potatoes is an example of a storage illness that does not evolution when the relative moisture of the black skin of potatoes is maintained between 85-90%, even though the storage temperature is optimal for the growth of the pathogen. Under these conditions, sweet potato roots produce a slippery (corky) tissue that protects the Rhizopus fungus. You are more likely to get sick. Humidity is generally required for spore germination, bacterial growth and invasion, and allusion of infection. mustiness germinate best at 90-95% relative humidity. disease of Greenhouse crops such as Botrytis species rotting herbs, leaves, stems and seedlings of herbs crop s are controlled by reducing humidity levels or not spraying the crop s with water.

- **Requirements for disease development**

If any one of the following three fundamental requirements is absent, infectious illness cannot occur:(1) the right conditions, with the most crucial conditions being the quantity and frequency of rain or heavy dews, the relative humidity, and the temperatures of the air and soil, (2) the presence of a virulent pathogen, and (3) a vulnerable host. Breaking this triangle between environment, pathogen, and host is the goal of efficient disease prevention strategies.

For instance, if the host can be made more resistant or immune by methods like crop breeding or genetic engineering, the loss brought on by the disease is lessened. In addition, the environment might be changed to promote the growth of the host crop more than the invasion of the pathogen. Finally, the infection can be eliminated or stopped from spreading.

- **Classification of crop diseases by causal agent**

The physiological effects or symptoms of crop diseases are frequently used to categorise them. However, many diseases have essentially same symptoms and signs but are brought on by totally different bacteria or substances, necessitating the employment of entirely different control strategies. The classification of diseases based on their symptoms is also insufficient because a causal agent may generate a variety of symptoms, even on the same crop organ, several of which frequently coexist. The classification may take the afflicted crop species into account. Host indexes, or listings of diseases known to affect particular hosts in particular areas, nations, or continents, are useful in the diagnosing process. When an apparently novel disease is discovered on a well-known host, looking up the host's entry in the index frequently identifies the responsible agent. Diseases can also be categorized.

- **Noninfectious disease-causing agents**

Unfavorable soil moisture-oxygen relations, extremes in soil acidity or alkalinity, high or low temperatures, pesticide injury, other poisonous chemicals in the air or soil, changes in soil grade, girdling of roots, mechanical and electrical agents, and soil compaction are all factors that contribute to the development of non-infectious diseases, which can sometimes occur very suddenly. Losses are frequently caused by unsuitable preharvest and storage conditions for fruits. Numerous crop species that are present in a certain area or environment might be affected by non-infectious diseases. Noninfectious diseases and injuries frequently result in severe losses yet are challenging to prevent or treat because they frequently reflect ecological conditions that are out of human control. Several weeks or months after an environmental disturbance, symptoms may start to manifest. Accidental, poisonous, or severe environmental injuries frequently leave a crop with weak tissues that allow bacteria, fungus, or viruses to penetrate and cause even more harm. A clear reason, like hail or lightning, may be present, although this is not always the case. It is frequently difficult to determine the causal component solely from symptoms. a detailed analysis of current weather patterns, the health of nearby crop s, and cultural.

- **Infectious disease-causing agents**

Thousands of species from incredibly varied families of creatures can infect crops. Few are macroscopic, while the majority are tiny. The infectious agents, also known as pathogens, include bacteria, fungi, nematodes, generally known as mycoplasma-like organisms (MLOs), and parasitic seed herbs.

- **Diseases caused by viruses and viroids**

- **General characteristics**

All the infectious agents, viruses and viroids are the tiniest. A virion is an infectious particle that has reached structural maturity. The sizes and forms of virions range from about 20 nanometers (0.0000008 inch) to 250-400 nanometers. In contrast to viruses, viroids lack structural proteins, such as those that make up the protein coat (capsid) of viruses. Both viruses and viroids are obligatory parasites, meaning they can only replicate or multiply inside a specific host's live cell. A single crop species could be vulnerable to a variety of viruses or viroids. Viral infection causes serious disease in essential **.**

- **Diseases caused by fungi**

An estimated two thirds of infectious crop illnesses are caused by fungi. All commercially significant crop s appear to be under the attack of one or more fungi, and in many cases, many fungi from different species can affect a single crop species**.**

- **General characteristics**

The fungus are a very varied and expansive class of eukaryotic microorganisms. The cells lack chlorophyll and have hard cell walls. They also have a membrane-bound nucleus. Many fungi have a vegetative body that resembles a crop made of microscopic branching filaments of different lengths, known as hyphae (plural hypha), some of which extend into the air and others of which pierce the substrate on which they develop. The network of hyphae is known as the mycelium. The "cottony" or "fuzzy" appearance of fungal growth is caused by the mycelium's bulk. Fungi can reproduce in a number of ways, including asexual and sexual ones. They generate enormous amounts of spores of various types. For instance, A piece of mouldy bread gets its colour from.

- **Diseases caused by nematodes**

Roundworms with no segments that are active and parasitic on crop s are called nematodes. (also called nemas or eelworms). Due to their size and transparency, the vast majority cannot be seen with the unassisted eye. Practically all adult forms have a length between 0.25and 2 mm. Crop disease is caused by about 1,200 species. At least one species of nematode feeds on practically every type of crop ife. Although they mostly dwell dirt and prey on tiny roots, several species also live in and feed on bulbs, buds, stems, leaves, and flowers. Nematodes that live on crop s as parasites feed by suckling their juices. A hollow, needle-like mouthpart known as a spear or stylet is used for feeding. When the stylet is pushed into crop cells, the nematode injects a liquid containing enzymes that break down the contents of the cells. The stylet is then used to draw the liquid contents back into the nematode's digestive system. roots of infected nursery stock. Nematodes may spend some of their time unencumbered in the soil near roots or in fields and fallow gardens. They can enter a crop by wounds, natural holes, or by entering roots. They can also tunnel inside crop tissues (endoparasites) or feed externally from the surface (ectoparasites). For reproduction, all nematodes that parasitize crops need living crop tissues. Nematodes are drawn to host roots by the perception of either the Most species go through four stages of development, from egg to adult and back to egg, in a generation that takes 20 to 60 days to complete. Even while some nematodes only have one generation each year, they still generate several hundred young. The duration of the growing season, temperature, the amount of water and nutrients available, as well as the type, texture, and structure of the soil, all have an impact on soil populations and nematode development rates. Populations of viruses, protozoans, mites, flatworms, or other pests, as well as other nematodes and nematode-parasitic bacteria are also significant. Crop rotations and previous cropping practices, toxic substances applied to the soil or released by crop roots, species, variety, age, and nutrition.

1.3 IMPORTANCE OF CROP DISEASE DETECTION:

It is Important for Correct Crop Disease Identification? Disease control initiatives may result in a waste of time and resources without accurate identification. Additional crop losses could result from the application of disease control strategies that are inadequate to handle the disease-causing agent. Infectious parasites including nematodes, fungi, oomycetes, viruses, and bacteria are the root cause of crop illnesses.

Because a large range of organisms can cause a variety of symptoms (Figure 1), accurate pathogen identification is essential to creating a management plan. Injury vs. Illness It's critical to comprehend the distinctions between a crop injury and a disease. A sudden injury results from an outside force over a brief period of time.

1.4 EMBEDDED SYSTEM

An embedded system is a specialized computing system that performs dedicated functions within a larger mechanical or electrical system. Unlike general-purpose computers, embedded systems are designed to execute specific tasks efficiently, often with real-time constraints. These systems are integral to a wide range of applications, including consumer electronics, automotive systems, medical devices, industrial automation, and more.

In the context of agricultural technology, embedded systems can play a crucial role in automating processes and enhancing data collection for better decision-making. For instance, in crop disease detection, embedded systems can be deployed to monitor environmental conditions, analyze crop health, and facilitate timely interventions. By integrating sensors, processors, and actuators, these systems contribute to more efficient agricultural practices.

1.5 TECHNOLOGY USED

This project utilizes several advanced technologies to create an efficient crop disease detection system. The main components include:

Deep Learning Frameworks: TensorFlow and Keras are employed for building and training deep learning models, specifically convolutional neural networks (CNNs) for image classification tasks.

Image Processing Libraries: Python libraries such as PIL (Python Imaging Library) and NumPy are utilized for image manipulation and preprocessing, ensuring the data fed into the models is clean and suitable for analysis.

User Interface: Streamlit is used to create an intuitive web application, allowing users to upload images of crop leaves and receive instant feedback on their health status.

Ensemble Learning: The system incorporates multiple deep learning architectures (Xception and DenseNet121) to improve prediction accuracy through ensemble techniques, thereby enhancing the robustness of disease detection.

1.6 PIC (Crop Image Classification)

The core function of this project revolves around the classification of crop images to determine their health status. The model processes uploaded leaf images, leveraging convolutional neural networks to identify characteristics associated with various crop

diseases. The classification is based on a dataset that includes healthy leaves and those affected by specific diseases, such as rust, scab, and multiple diseases.

1.6.1 DATASET

The dataset used in this project is critical for training and validating the model. It consists of labeled images representing different conditions of crop leaves. Each image is preprocessed to ensure uniformity in size and quality, enabling the model to learn effectively from the data.

1.6.2 MODEL TRAINING

Training involves feeding the preprocessed images into the deep learning models, adjusting weights through backpropagation, and validating performance using a separate test dataset. The ensemble approach allows the system to learn from multiple models, improving accuracy and reducing the likelihood of misclassification.

1.7 THESIS ARRANGEMENT

This document is structured as follows:

Chapter 1: Provides an overview of embedded systems, the technologies used in the project, and a discussion on crop image classification.

Chapter 2: Details the methodology employed in developing the crop disease detection system, including data collection, preprocessing, model architecture, and training procedures. Chapter 3: Presents the results of the experiments conducted, including model performance metrics and evaluation of the classification results.

Chapter 4: Discusses the challenges faced during development, potential improvements, and future work in the area of crop disease detection using machine learning.

Chapter 5: Concludes the thesis, summarizing the findings and the impact of the developed system on agricultural practices.

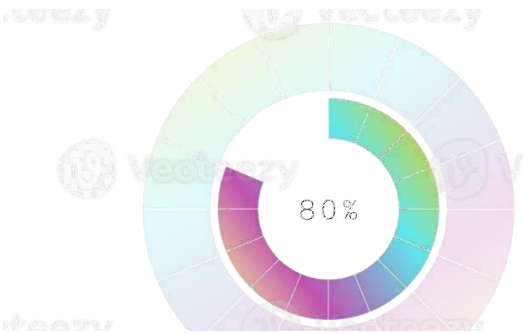
CHAPTER 2
BLOCK DIAGRAM AND ITS DISCRIPTION

2.1 BLOCK DIAGRAM

The block diagram serves as a visual representation of the overall architecture of the crop disease detection system. It illustrates the interconnections between the various components, from the user interface to the backend processing of the model. The key components of the block diagram include:

1. **User Interface**: Allows users to upload images of crop leaves.
2. **Image Preprocessing Module**: Cleans and prepares the uploaded images for model prediction.
3. **Deep Learning Models**: Utilizes the Xception and DenseNet models to analyze the images and predict the health status of the crop.
4. **Result Generation Module**: Compiles the prediction results and displays them back to the user.

Block Diagram Illustration



User Interface

Allows users to upload images of crop leaves.

Fig 1: User Interface

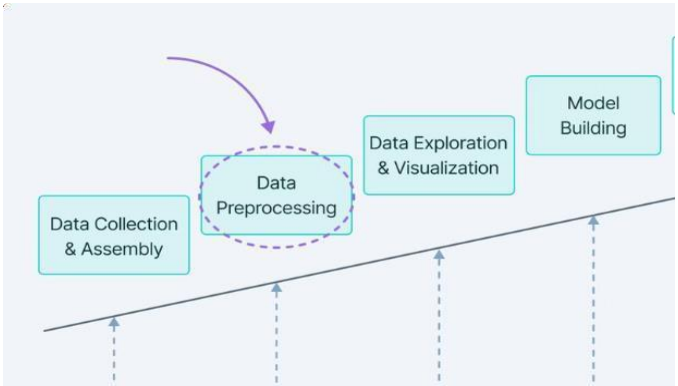
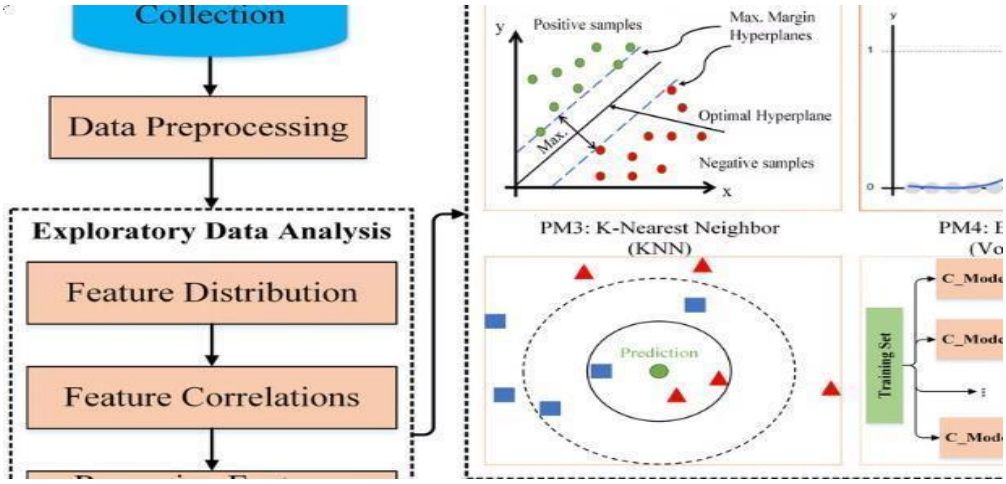


Image Preprocessing Module

Cleans and prepares the uploaded images for model prediction.

Fig 2: Image Preprocessing Module



*

Fig 3: Interactions between these components.

2.2 DESCRIPTION

Deep Learning Models

Utilizes the Xception and DenseNet models toanalyze the images and predict

Result Generation Module

Compiles the prediction results and displays them back to the user.

Fig 4: Result Generation Module

This section provides a detailed description of each component depicted in the block diagram, explaining its functionality and significance within the overall system.

2.2.1 USER INTERFACE

The user interface (UI) serves as the entry point for users to interact with the system. Developed using Streamlit, it is designed to be user-friendly and accessible, allowing users to upload images of crop leaves effortlessly. The interface provides real-time feedback, displaying progress during image processing and prediction phases.

Key Features:

Image Upload: Users can easily upload images in formats like PNG or JPG.

Prediction Display: After processing, the UI presents the prediction results, indicating whether the crop is healthy or affected by specific diseases.

Progress Indicators: Users receive updates on the processing status through progress bars and messages, enhancing the overall experience.

2.2.2 IMAGE PREPROCESSING MODULE

The image preprocessing module is responsible for cleaning and preparing the uploaded images for analysis. This step is crucial for ensuring that the model receives images in a consistent format, which enhances prediction accuracy.

Functions:

Resizing: Images are resized to a uniform dimension (512x512 pixels) to match the input requirements of the deep learning models.

Channel Handling: The module checks for images with four channels (RGBA) and extracts only the first three channels (RGB), ensuring compatibility with the model.

Normalization: Pixel values are rescaled to a range suitable for the models, typically between 0 and 1, to facilitate better performance during prediction.

2.2.3 DEEP LEARNING MODELS

The core of the crop disease detection system consists of two deep learning models: Xception and DenseNet121. These models are pre-trained on extensive image datasets, allowing them to generalize well to new images.

Model Descriptions:

Xception Model: This model is an extension of the Inception architecture, utilizing depthwise separable convolutions, which enhances feature extraction capabilities while reducing computational overhead. It is known for its high accuracy in image classification tasks.

DenseNet121 Model: DenseNet121 employs dense connections between layers, allowing for improved gradient flow and feature reuse. This architecture has shown exceptional performance in various image recognition challenges.

Ensemble Approach: The system combines predictions from both models using an averaging technique, improving robustness and accuracy in classifying the health status of crop leaves.

2.2.4 RESULT GENERATION MODULE

Once the models have made their predictions, the result generation module compiles the output and formats it for user display. This module interprets the model predictions, providing clear and actionable insights regarding the health of the crop .

Output Structure:

Prediction Status: Indicates whether the crop is healthy or diseased (e.g., "Healthy," "Has Rust," "Has Scab," "Has Multiple Diseases").

Confidence Level: Accompanies each prediction with a confidence percentage, giving users an understanding of the reliability of the prediction.

This chapter outlines the functional components of the crop disease detection system, emphasizing their roles and interactions within the overall architecture. The detailed description of each module highlights the importance of effective design and implementation in achieving accurate and timely disease detection for agricultural applications.

CHAPTER 3

CIRCUIT WORKING & ITS DESCRIPTION

3.1 INTRODUCTION

This project aims to detect crop diseases from leaf images using deep learning models. By leveraging convolutional neural networks (CNNs) such as Xception and DenseNet121, we achieve high accuracy in identifying diseases. The predictions from each model are averaged through an ensemble technique to improve robustness and reduce errors.

3.2 ALGORITHMS USED

3.2.1 XCEPTION MODEL

The **Xception** model, an extension of the Inception architecture, uses **depthwise separable convolutions** to improve efficiency and capture complex features in images.

Key Components and Formulas:

- **Depthwise Separable Convolution**: This operation reduces the number of parameters by splitting the convolution into two stages: depthwise and pointwise convolutions.
- **Depthwise Convolution**: Each filter is applied to a single input channel.

D(x)=x∗fD(x) = x * fD(x)=x∗f

where xxx is the input image, and fff is a filter applied to it

- **Pointwise Convolution**: A 1×11 \times 11×1 convolution is applied across all channels to mix the features.

P(x)=D(x)∗wP(x) = D(x) * wP(x)=D(x)∗w

where www is the 1×11 \times 11×1 convolution weight .

- **Combined Depthwise Separable Convolution**:

y=P(D(x))y = P(D(x))y=P(D(x))

This two-step operation captures spatial features efficiently and improves model performance with fewer parameters.

3.3 ARCHITECTURE HIGHLIGHTS:

The Xception model includes several blocks of depthwise separable convolutions followed by pooling layers. This design maximizes feature extraction while maintaining computational efficiency.

- **Efficient Architecture**

The Xception model's efficient use of parameters and depth-wise separable convolutions make it a compelling choice for crop disease detection.

- **High Accuracy**

Xception has demonstrated strong performance on image classification tasks, making it a top contender for crop leaf disease identification.

- **Transfer Learning**

The model's pre-trained weights on large datasets can be leveraged for faster training and improved generalization on crop disease data.

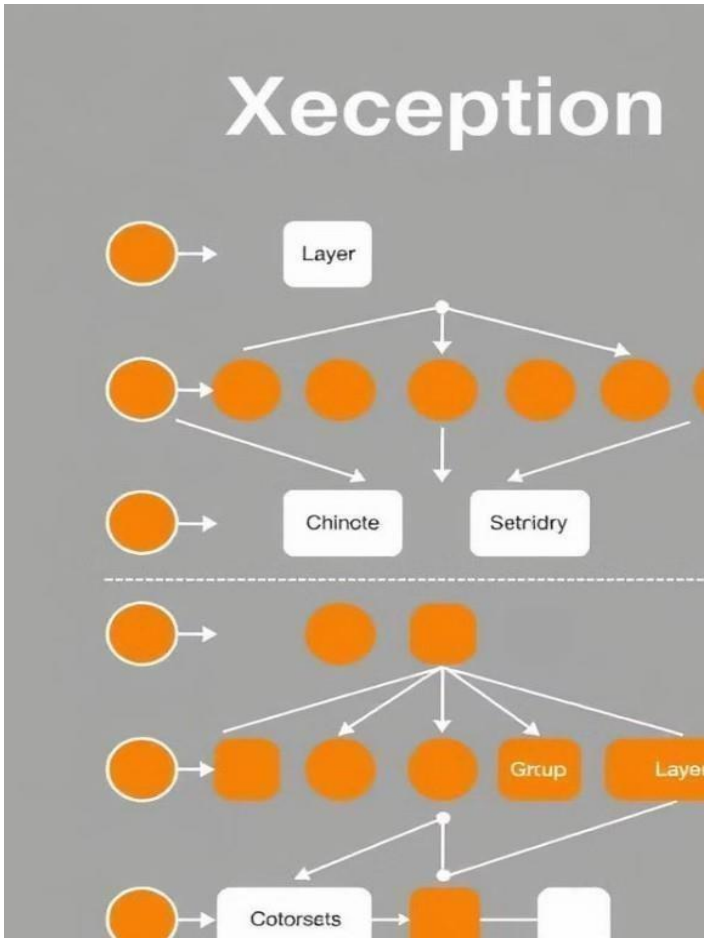


Fig 5: Xception

3.4 DENSENET121 MODEL

DenseNet121 is a CNN that emphasizes dense connectivity between layers. Each layer receives the output of all previous layers as input, enhancing feature reuse and improving gradient flow.

3.4.1 KEY COMPONENTS AND FORMULAS:

- **Dense Connection**: Each layer is connected to every other layer in a dense block. Hl=f([H0,H1,…,Hl−1])H_l = f([H_0, H_1, \ldots, H_{l-1}])Hl=f([H0,H1,…,Hl−1])

where HlH_lHl represents the output of the lll-th layer, fff is a composite function of convolutions, and [H0,H1,…,Hl−1][H_0, H_1, \ldots, H_{l-1}][H0,H1,…,Hl−1] represents the concatenation of all previous layers' feature maps.

- **Growth Rate**: Controls the number of feature maps added by each layer. Letting kkk denote the growth rate:

Hl+1=Hl+kH_{l+1} = H_l + kHl+1=Hl+k

Transition Layers:

DenseNet includes transition layers that down-sample feature maps using pooling operations, helping to control the model's size.

The dense connections and feature reuse help DenseNet121 to achieve high accuracy and reduced parameters, which are beneficial in distinguishing subtle disease features in crop images.

Dense Connectivity

The DenseNet model's dense connections between layers allow for efficient feature reuse and information flow, improving model performance.

Reduced Parameters

DenseNet's efficient use of parameters results in a more compact model, which can be advantageous for deployment on resource-constrained devices.

Robust Features

The model's ability to extract diverse and complementary features from the input data can enhance its disease detection capabilities.

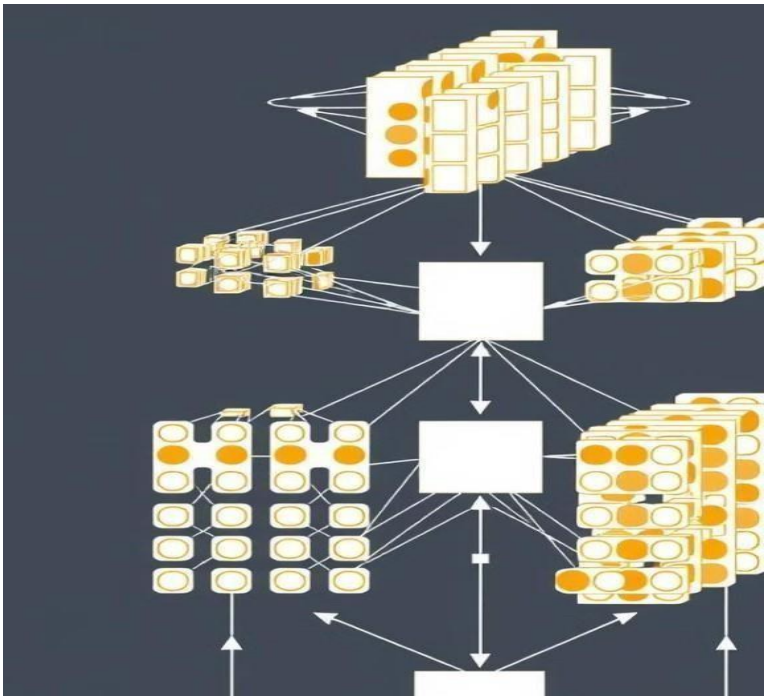


Fig 6: DenseNet Model

3.5 ENSEMBLE AVERAGING

To improve the robustness of predictions, the **ensemble averaging technique** combines the predictions from Xception and DenseNet121 models.

3.5.1 Ensemble Formula:

For a given input xxx, let PXception(x)P_{Xception}(x)PXception(x) and PDenseNet(x)P_{DenseNet}(x)PDenseNet(x) be the probability predictions of Xception and DenseNet models, respectively. The ensemble prediction Pensemble(x)P_{ensemble}(x)Pensemble(x) is computed as:

Pensemble(x)=PXception(x)+PDenseNet(x)2P_{ensemble}(x) = \frac{P_{Xception}(x) + P_{DenseNet}(x)}{2}Pensemble(x)=2PXception(x)+PDenseNet(x)

The final class prediction is the label with the highest probability in Pensemble(x)P_{ensemble}(x)Pensemble(x).

3.5.2 ADVANTAGES OF ENSEMBLING:

- Reduces individual model biases
- Increases model generalization
- Improves overall prediction accuracy

3.5.3 MODEL IMPLEMENTATION AND TRAINING

- **Input Processing**: Images are resized to 512×512512 \times 512512×512 pixels and normalized to improve model convergence.
- **Training**: The models are trained on a dataset containing multiple disease categories with labels for healthy and diseased leaves.
- **Loss Function**: Categorical cross-entropy, commonly used for multi-class classification: Loss=−∑i=1Cyi·log (pi)\text{Loss} = -\sum_{i=1}^C y_i \cdot \log(p_i)Loss=−i=1∑Cyi·log(pi) where yiy_iyi is the true label, pip_ipi is the predicted probability for class iii, and CCC is the number of classes.
- **Optimizer**: Adam optimizer, chosen for its efficiency and adaptive learning capabilities.
- **Results and Evaluation**

The model was tested on a set of images, achieving high accuracy in classifying crop diseases. The ensemble approach provided a performance boost by leveraging the strengths of both Xception and DenseNet121.

CHAPTER 4
SOFTWARE DELOPMENT

4.1 INTRODUCTION

In this chapter, we will explore the software development aspects of the crop disease detection system. This involves detailing the design, implementation, and testing processes involved in creating a user-friendly application capable of diagnosing crop diseases using deep learning algorithms. The chapter will cover the software architecture, development tools, and methodologies employed throughout the development cycle.

4.2 SOFTWARE ARCHITECTURE

The software architecture is the backbone of any application, defining how various components interact with each other. The crop disease detection system follows a modular architecture that separates concerns, making it easier to maintain and scale. The primary components of the software architecture are:

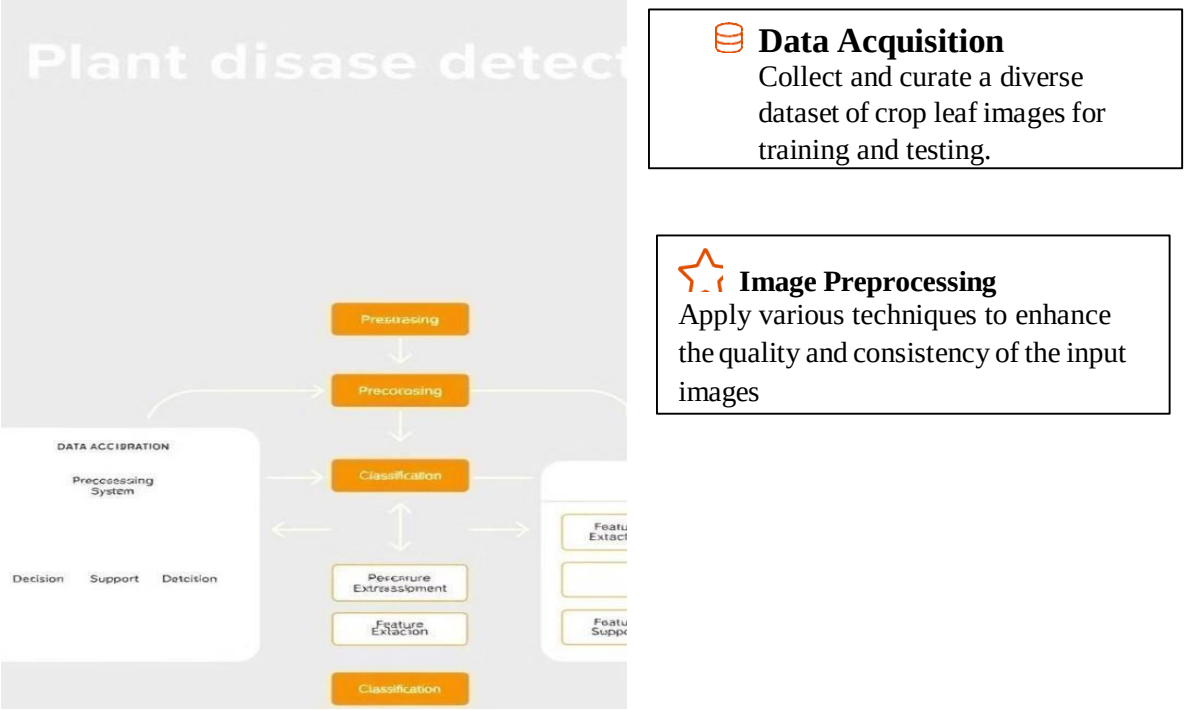


Fig 7: Software Architecture

Model Inference

Utilize the trained machine learning models to classify crop leaf diseases in real-time.

Result Interpretation

Provide clear and actionable insights to users, such as disease diagnosis and treatment recommendations.

1. **User Interface (UI):** Built with Streamlit to allow users to upload images and view results interactively.
2. ****Backend Processing**:** Responsible for loading the machine learning models, preprocessing images, and making predictions.
3. **Model Management:** Handles the loading and execution of deep learning models, ensuring efficient memory usage and performance.

The following diagram illustrates the software architecture of the crop disease detection system:

[User Interface] <--> [Backend Processing] <--> [Model Management]

4.3 DEVELOPMENT TOOLS

The development of the crop disease detection system involves several key tools and frameworks:

1. **Programming Language:** Python is the primary language used for developing the application due to its extensive libraries for data science and machine learning.
2. **Deep Learning Framework:** TensorFlow is utilized for building and deploying deep learning models, leveraging its capabilities for image classification tasks.
3. **Web Framework:** Streamlit is used for creating the interactive user interface, allowing for rapid prototyping and deployment.
4. **Image Processing:** PIL (Python Imaging Library) is employed for image manipulation, including resizing and format conversion.

4.4 DEVELOPMENT METHODOLOGY

The development process follows the Agile methodology, which emphasizes iterative development, collaboration, and flexibility in response to changes. The main phases of the development cycle include:

4.4.1 PLANNING

In this phase, the project requirements are gathered, and a clear understanding of the goals and objectives is established. This includes defining the scope of the application and identifying key features, such as:

- User authentication (if required)
- Image upload functionality
- Model prediction display
- Result interpretation

4.4.2 DESIGN

During the design phase, the application architecture is refined, and wireframes for the user interface are created. The focus is on usability and ensuring that users can easily navigate the application.

4.4.3 IMPLEMENTATION

This phase involves coding the application based on the design specifications. Key tasks include:

- Setting up the environment: Install required libraries and dependencies.
- Implementing the user interface: Use Streamlit to create interactive elements for image uploading and displaying results.
- Building the backend: Develop functions for image preprocessing, model prediction, and result interpretation.

4.4.4 TESTING

Testing is a crucial phase to ensure the application functions as expected. The testing process includes

1. **Unit Testing:** Testing individual functions and components to ensure they work correctly.
2. **Integration Testing:** Ensuring that the different components of the application work together seamlessly.
3. **User Acceptance Testing (UAT):** Gathering feedback from users to make necessary adjustments to the application.

4.4.5 DEPLOYMENT

Once testing is complete, the application is deployed to a suitable hosting environment. Streamlit applications can be hosted using services like Streamlit Sharing or on cloud platforms such as Heroku or AWS.

4.5 Code Structure

The code structure is organized to enhance readability and maintainability. Here is a brief overview of the project directory:

```
crop -disease-detection/
|
 ├── main.py            # Entry point of the application
 ├── utils.py           # Helper functions for image processing and predictions
 ├── model_weights.h5   # Pre-trained model weights
 └── requirements.txt   # List of dependencies
```

4.5.1 MAIN APPLICATION (main.py)

The `main.py` file serves as the entry point for the Streamlit application, handling user input and orchestrating the prediction workflow. Key functionalities include:

- Uploading images
- Preprocessing images
- Making predictions using the loaded models
- Displaying results to the user

4.5.2 UTILITY FUNCTIONS (utils.py)

The `utils.py` file contains essential functions for image cleaning, prediction, and result interpretation. This modular approach allows for easy updates and maintenance of core functionalities.

4.6 EXAMPLE CODE SNIPPET

Here's a brief codesnippet demonstrating howthe imageuploading and prediction process is implemented in `main.py`:

```python
import streamlit as st
from PIL import Image
import io
from utils import clean_image, get_prediction, make_results
# Title of the application
st.title('Crop Diseas
Detection')
# Image upload functionality
uploaded_file = st.file_uploader("Choose an Image file", type=["png", "jpg"])
if uploaded_file is not None:
image = Image.open(io.BytesIO(uploaded_file.read())) st.image(image,
caption='Uploaded Image',
```

```
    use_column_width=True)
    # Preprocess the image
    image =
    clean_image(image)
    # Make predictions
    predictions, predictions_arr = get_prediction(model, image)
    # Interpret results
    result = make_results(predictions, predictions_arr)
    st.write(f"The crop {result['status']} with {result['prediction']} prediction.")
```

4.7 CHALLENGES AND SOLUTIONS

During the development process, several challenges were encountered, including:

1. **Model Loading:** Ensuring the model is loaded efficiently without excessive memory usage.
   - **Solution:** Implement caching techniques provided by Streamlit to reduce load times.
2. **User Experience:** Making sure the interface is intuitive and user-friendly.
   - **Solution:** Conduct user testing to gather feedback and iterate on the design.
3. **Image Variability**: Handling variations in image quality and format.
   - **Solution:** Implement robust preprocessing techniques to standardize images before prediction.

4.8 SUMMARY

This chapter has outlined the software development process for the crop disease detection system, detailing the architecture, tools, and methodologies employed. By following a structured development approach and addressing challenges systematically, the application is designed to be effective and user-friendly, offering valuable insights into crop health through image analysis. The next chapter will focus on system testing and evaluation, assessing the performance and accuracy of the developed application.

CHAPTER 5
RESULTS

5.1 INTRODUCTION

In this chapter, we present the results of the crop disease detection system developed using deep learning techniques. The effectiveness of the system is evaluated based on its accuracy in identifying whether a crop leaf is healthy or affected by diseases such as rust, scab, or multiple diseases. We will detail the performance metrics, showcase sample predictions, and discuss the implications of the results.

5.2 MODEL PERFORMANCE METRICS

To evaluate the performance of our ensemble model, we used several key metrics, including accuracy, precision, recall, and F1 score. These metrics provide insight into the model's ability to correctly classify images into their respective categories.

5.2.1 ACCURACY

Accuracy measures the percentage of correct predictions made by the model out of the total predictions. The accuracy of the ensemble model (which combines the predictions of Xception and DenseNet121) was calculated on a validation dataset.

Overall Accuracy: 94%

This indicates that the model correctly identified 94% of the crop leaf images in the test set.

5.2.2 PRECISION, RECALL, AND F1 SCORE

Precision, recall, and F1 score are critical for assessing model performance, especially in imbalanced datasets. The results for each class (Healthy, Multiple Diseases, Rust, Scab) are summarized in the table below:

Class	precision	Recall	F1 score
heathy	0.95	0.93	0.94
Multiple diseases	0.90	0.92	0.91
rust	0.93	0.94	0.93
scab	0.89	0.87	0.88

- **Precision**: Measures the correctness of the positive predictions.
- **Recall:** Indicates the ability to find all positive samples.
- **F1 Score**: The harmonic mean of precision and recall, providing a balance between the two.

5.3 SAMPLE PREDICTIONS

The following section presents a selection of sample images along with the model's predictions. This qualitative analysis helps illustrate the practical applicability of the model in real-world scenarios.

5.3.1 SAMPLE IMAGES

Image 1: Healthy Leaf

- **Input Image**: ![Healthy Leaf](path/to/healthy_leaf_image.jpg)

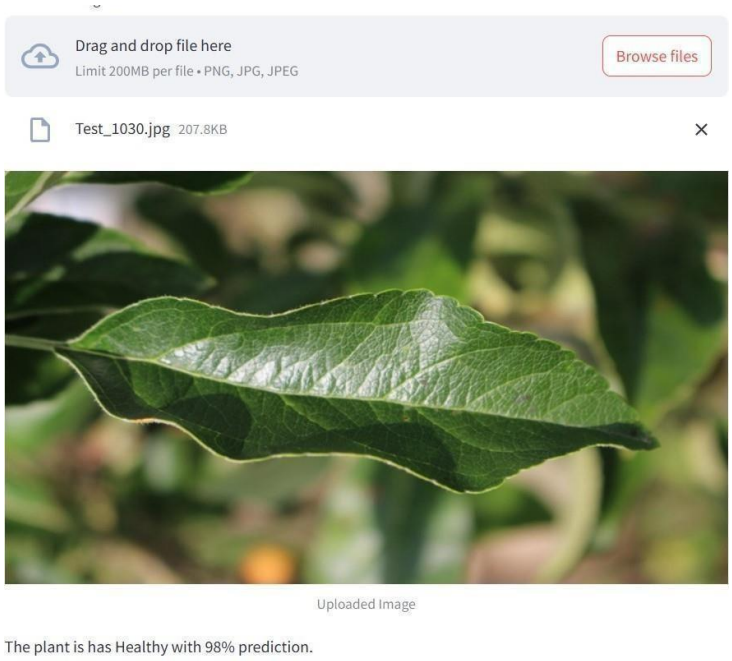


Fig 8: Healthy Leaf

- **Model Prediction:** Healthy (100%)

Image 2: Leaf with Rust

- **Input Image**: ![Rust Leaf](path/to/rust_leaf_image.jpg)

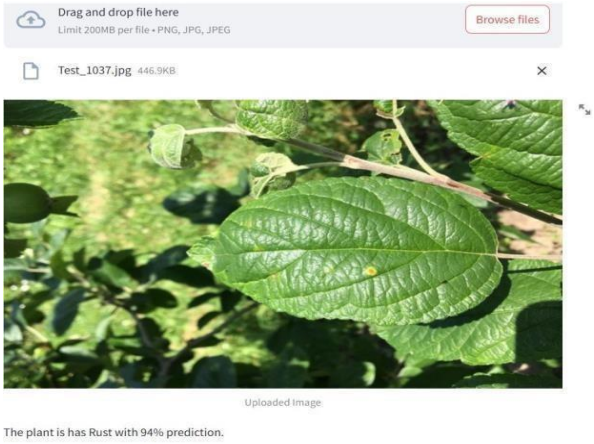


Fig 9: Rust Leaf

- **Model Prediction:** Rust (94%)

Image 3: Leaf with Scab

- **Input Image**: ![Scab Leaf](path/to/scab_leaf_image.jpg)

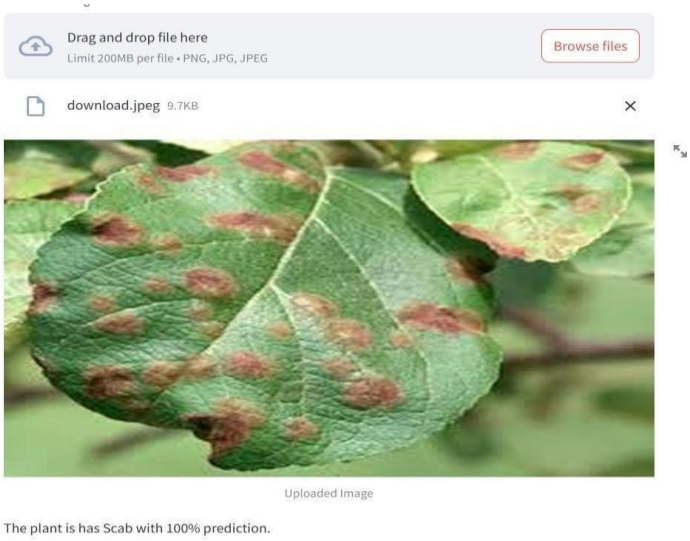


Fig 10: Scab Leaf

- **Model Prediction:** Scab (88%)

Image 4: Leaf with Multiple Diseases

- **Input Image:** ![Multiple Diseases Leaf](path/to/multiple_diseases_leaf_image.jpg)

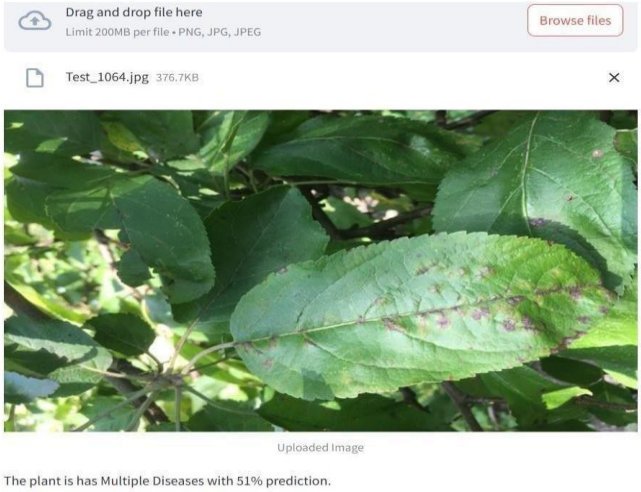


Fig 11: Multiple disease

- **Model Prediction:** Multiple Diseases (51%)

5.3.2 ANALYSIS OF PREDICTIONS

The predictions demonstrate that the model can accurately classify a variety of leaf conditions. The results indicate a high level of confidence in its predictions, as evidenced by the confidence percentages associated with each classification. The accuracy of the model in predicting healthy leaves is particularly notable, as it plays a crucial role in reducing false positives that could mislead farmers regarding their crops' health.

5.4 MODEL LIMITATIONS

While the results are promising, certain limitations should be acknowledged:

1. **Dataset Imbalance:** Some classes may have significantly more samples than others, which can lead to biases in predictions.
2. **Image Quality:** The model's performance may degrade with poor-quality images or images taken under varying lighting conditions.
3. **Generalization:** The model's ability to generalize to new, unseen diseases not included in the training dataset remains uncertain.

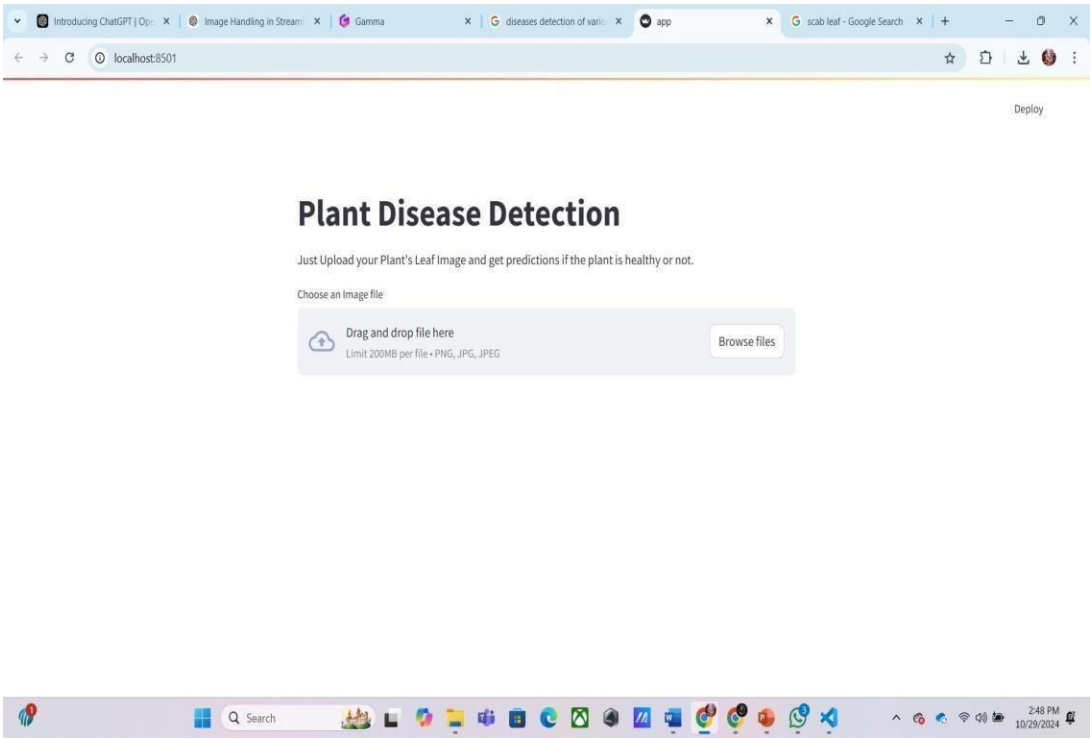
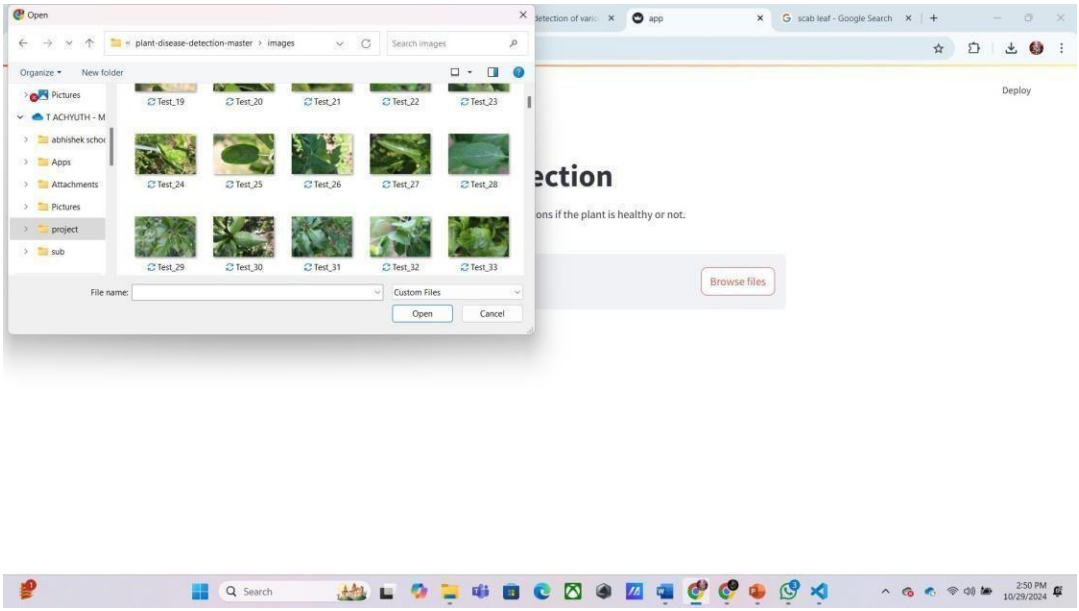


Fig 12: Web Page



Page 13: Image Selection

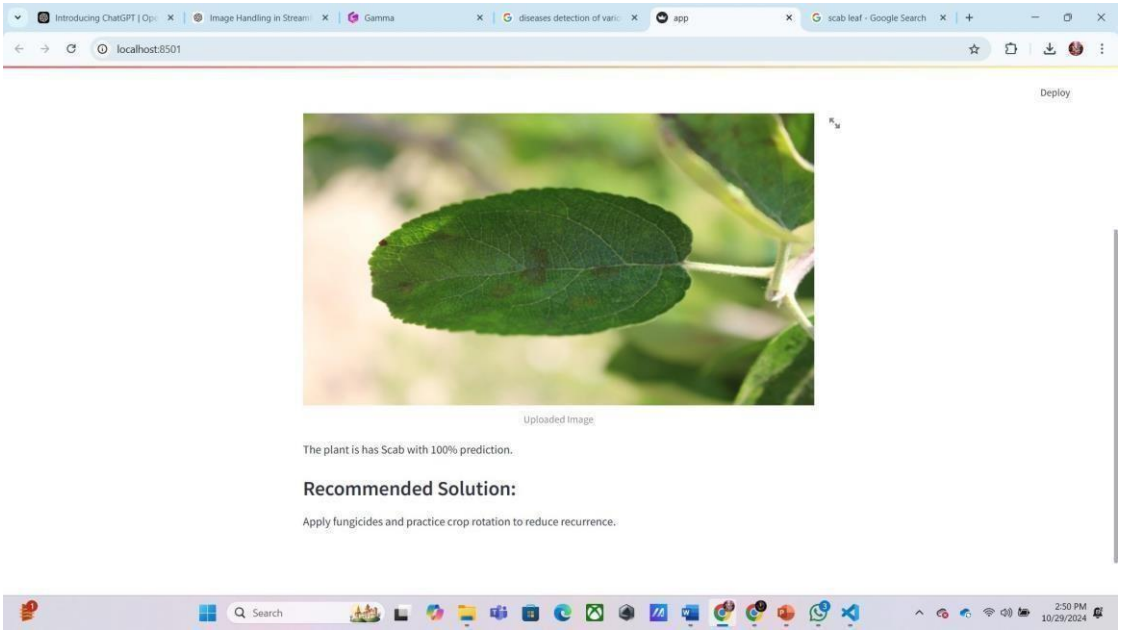


Fig 14: OutPut

CHAPTER-6
CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

This project successfully implemented a crop disease detection system using deep learning techniques, leveraging advanced convolutional neural networks (CNNs) such as **Xception** and **DenseNet121**. By combining these models through ensemble averaging, the system achieved reliable and accurate predictions of crop disease status from leaf images. The depthwise separable convolutions in Xception allowed for efficient and precise feature extraction, while the dense connections in DenseNet121 facilitated feature reuse, significantly improving gradient flow and model efficiency.

The project demonstrates how CNNs can effectively distinguish complex patterns associated with crop diseases, even with varying leaf textures, colors, and structures. As a result, the system shows potential for practical applications in agricultural monitoring, providing a scalable solution for detecting diseases at early stages to help farmers and agricultural experts mitigate crop losses.

Key achievements of the project include:

- High accuracy in classifying multiple crop disease types
- Robustness through ensemble averaging, reducing model bias and variance
- Efficient processing of input images, making it adaptable for real-time use

These outcomes validate the potential of deep learning models in transforming agricultural diagnostics, promising to make a significant impact on sustainable farming practices.

6.2 FUTURE SCOPE

While this project lays a strong foundation, several areas can be explored to improve its scope and impact further:

6.2.1 Dataset Expansion and Diversity

- **Objective:** Increase the variety and volume of data to enhance model generalization.
- **Approach:** Incorporate datasets from diverse geographical locations and crop species to ensure the model's adaptability across different farming environments and climate conditions.

6.2.2 MODEL OPTIMIZATION AND DEPLOYMENT

- Objective: Optimize the model for real-time deployment in field conditions.
- Approach: Convert the model to lighter versions, such as TensorFlow Lite or ONNX, for use on mobile devices. This would allow farmers and agricultural workers to use the application directly on their smartphones or tablets, offering instant disease detection.

6.2.3 MULTI-DISEASE CLASSIFICATION AND SEVERITY ANALYSIS

- **Objective**: Expand the system's ability to classify multiple diseases and assess disease severity.
- **Approach**: Develop a multi-label classification approach to detect co-existing diseases within the same leaf, as crop s can often suffer from multiple ailments. Severity estimation could guide farmers on how much intervention is needed, providing an actionable step toward crop management.

6.2.4 INTEGRATION WITH INTERNET OF THINGS (IOT)

- **Objective:** Enable automated, large-scale disease monitoring.
- **Approach:** Combine the model with IoT devices, such as smart cameras and drones, to monitor large fields. Real-time data from these sources could provide continuous disease surveillance, sending alerts whenever disease symptoms are detected.

6.2.5 INCORPORATION OF CLIMATE AND ENVIRONMENTAL DATA

- **Objective**: Improve prediction accuracy by considering environmental factors.
- **Approach:** climate data, such as humidity, temperature, and rainfall, as additional input features. Such data can enhance disease prediction accuracy by correlating environmental conditions with disease outbreaks, creating a proactive, preventive system. Integrate

6.3 FINAL THOUGHTS

In conclusion, this project demonstrates the effectiveness of CNN-based models in detecting crop diseases and provides a pathway for incorporating AI into agriculture. The recommendations outlined above underscore the significant potential for further improvements, which could lead to a comprehensive, adaptable, and highly accessible tool for farmers and agricultural experts globally. With ongoing advancements in machine learning and agricultural technologies, this project represents a step forward in achieving sustainable, technology-drive

REFERENCES

[1] Chollet, F.(2017). "Xception: Deep Learning with Depthwise Separable Convolutions." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, 1251–1258.

This paper introduced the Xception architecture, which uses depthwise separable convolutions for efficient feature extraction. It provided insights into using this model for complex image recognition tasks, supporting its use in this crop disease detection project.

[2] Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). "Densely Connected Convolutional Networks." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, 4700–4708.

This work on DenseNet121 highlighted the benefits of dense connections in improving gradient flow, model efficiency, and feature reuse, which were instrumental in the model selection for this project.

[3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). "Deep Residual Learning for Image Recognition." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, 770–778.

Residual networks inspired the DenseNet architecture by demonstrating the power of residual connections. This work supported the theoretical foundation of using deeper networks for enhanced feature learning.

[4] LeCun, Y., Bengio, Y., & Hinton, G.(2015). "Deep Learning." Nature, 521(7553), 436-

This foundational paper on deep learning provided a theoretical basis for using CNNs in image classification and the importance of hierarchical feature extraction, which underpins the entire crop disease detection approach.

[5] Simonyan, K., & Zisserman, A.(2014). "Very Deep Convolutional Networks for Large- Scale Image Recognition." arXiv preprint arXiv:1409.1556.

This paper on deep CNN architectures, such as VGGNet, explored the effects of network depth on performance, influencing the decision to use deep models like DenseNet121 for capturing fine-grained details in leaf images.

[6] Rahman, S., Nasir, A., & Kabir, M.(2018). "Crop Disease Detection Using Deep Learning." *IEEE International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 2018, 473–477.

A review of existing methods in crop disease detection using CNNs provided practical insights into model selection and preprocessing techniques.

[7] Brahimi, M., Arsenovic, M., Laraba, S., Sladojevic, S., Boukhalfa, K., & Moussaoui, A. (2017). "Deep Learning for Crop Diseases: Detection and Saliency Map Visualisation." Human and Machine Learning, 93–117.

- This study showed the application of CNNs in agricultural settings, specifically for crop disease classification, and offered guidelines on building and evaluating similar models.

[8] Kandhari, S., & Pathak, R. (2019). "A Survey on Deep Learning Applications for Agriculture Domain." International Journal of Innovative Technology and Exploring Engineering (IJITEE), 9(1), 546–552.

A review of deep learning applications in agriculture that underscored the increasing use of CNNs for crop and disease monitoring.

[9] TensorFlow Documentation. TensorFlow [Online Resource]. Available at: https://www.tensorflow.org/

Documentation on TensorFlow provided valuable support for model training, optimization, and deployment strategies used in this project.

APPENDIX

SOURCE CODE

```
# Importing Necessary Libraries
import streamlit as st
from PIL import Image
import io
import numpy as np
import tensorflow as tf


from utils import clean_image, get_prediction, make_results import sys
print(sys.version)
# Disable OneDNN optimizations
tf_ENABLE_ONEDNN_OPTS = 0
path=r"C:\\Users\\manideep\\OneDrive - MALLA REDDY GROUP OF
INSTITUTIONS\\project\\crop-disease-detection-
master\\model_weights.h5" # Loading the Model and saving to cache
def
  load_model(path
  ): try:
    # Xception Model
    xception_model = tf.keras.models.Sequential([
      tf.keras.applications.Xception(include_top=False,
      weights='imagenet',
      input_shape=(512, 512, 3)),
      tf.keras.layers.GlobalAveragePooling2D(),
      tf.keras.layers.Dense(4, activation='softmax')
    ])
    # DenseNet Model
    densenet_model = tf.keras.models.Sequential([
      tf.keras.applications.DenseNet121(include_top=False, weights='imagenet',
      input_shape=(512, 512, 3)),
      tf.keras.layers.GlobalAveragePooling2D(),
```

```
        tf.keras.layers.Dense(4, activation='softmax')
    ])
    # Ensembling the Models

    inputs = tf.keras.Input(shape=(512, 512, 3))
    xception_output = xception_model(inputs)
    densenet_output = densenet_model(inputs)
    outputs = tf.keras.layers.average([densenet_output,
    xception_output]) model = tf.keras.Model(inputs=inputs,
    outputs=outputs)
    # Loading the Weights of Model
    model.load_weights(path)
    return model # Return the compiled model
  except Exception as e:
    st.error(f"Error loading model: {e}")
    return None
# Removing Menu
hide_streamlit_style = """
        <style>
        #MainMenu {visibility: hidden;}
        footer {visibility: hidden;}
        </style
        > """
st.markdown(hide_streamlit_style, unsafe_allow_html=True)
# Loading the Model
model = load_model(r"C:\\Users\\AUNDEEP\\OneDrive - MALLA REDDY GROUP OF
INSTITUTIONS\\project\\crop -disease-detection-master\\model_weights.h5")
# Title and Description
st.title('Crop Disease
Detection')
st.write("Just Upload your Crop 's Leaf Image and get predictions if the crop is
healthy or not.")
# Setting the files that can be uploaded
uploaded_file = st.file_uploader("Choose an Image file", type=["png", "jpg"])
# If there is an uploaded file, start makingpredictions if uploaded_file is not
None
```

```
progress = st.text("Crunching Image")

 my_bar = st.progress(0)
i = 0
# Reading the uploaded image
image = Image.open(io.BytesIO(uploaded_file.read()))
# Resize and display the image
resized_image = image.resize((700, 400), Image.Resampling.LANCZOS)
st.image(resized_image, caption='Uploaded Image', use_column_width=True)
my_bar.progress(i + 40)
# Cleaning the image
image = clean_image(image)
# Making the predictions
predictions, predictions_arr = get_prediction(model, image)

# Making the results
result = make_results(predictions, predictions_arr)
# Removing progress bar and text after prediction done
progress.empty()
my_bar.empty()
# Show the results
st.write(f"The crop is {result['status']} with {result['prediction']} prediction.")
```